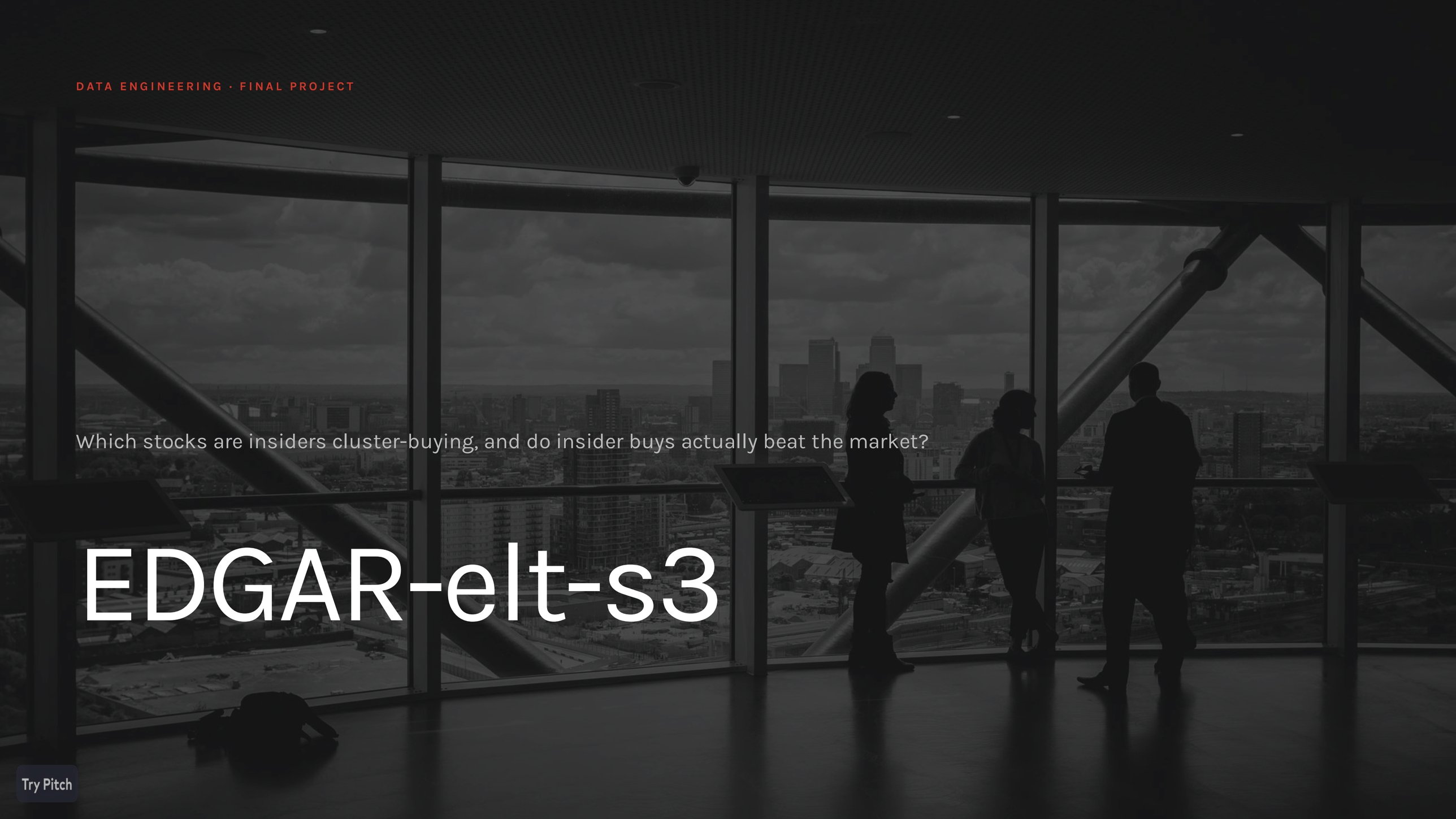
DATA ENGINEERING · FINAL PROJECT



Which stocks are insiders cluster-buying, and do insider buys actually beat the market?

EDGAR-elt-s3

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01 · THE DATA

SEC EDGAR

What it is

Insider trades (Forms 3, 4, 5), financial statements (10-K, 10-Q), CIK ↔ ticker list, and daily stock prices.

Where it came from

SEC insider transaction and financial statement data sets, SEC company_tickers_exchange.json, and Yahoo Finance via yfinance.

License and update frequency

SEC data is public domain, accessed per SEC fair access policy (declared User-Agent, max 10 req/s). Filings update quarterly; tickers and prices daily.

Why I picked it

Insiders know their companies best. ~20 GB of real, messy, dated filings make it possible to test whether following them pays off.

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What I set out to answer

“By the end of this, I'll be able to answer who is cluster-buying and whether it pays, about insider trading in US stocks.”

QUESTION 01

Which stocks are insiders cluster-buying right now?

QUESTION 02

Do insider buys actually beat the market?

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Five choices, wired end to end

① SOURCE

SEC EDGAR + yfinance

Forms 3/4/5, 10-K/10-Q, tickers, daily prices

② RAW LANDING

RustFS (S3)

Raw files saved untouched, replayable

③ ANALYTICAL STORE

ClickHouse

Bronze → silver → gold, plus dbt data tests

⑤ SERVING

Streamlit

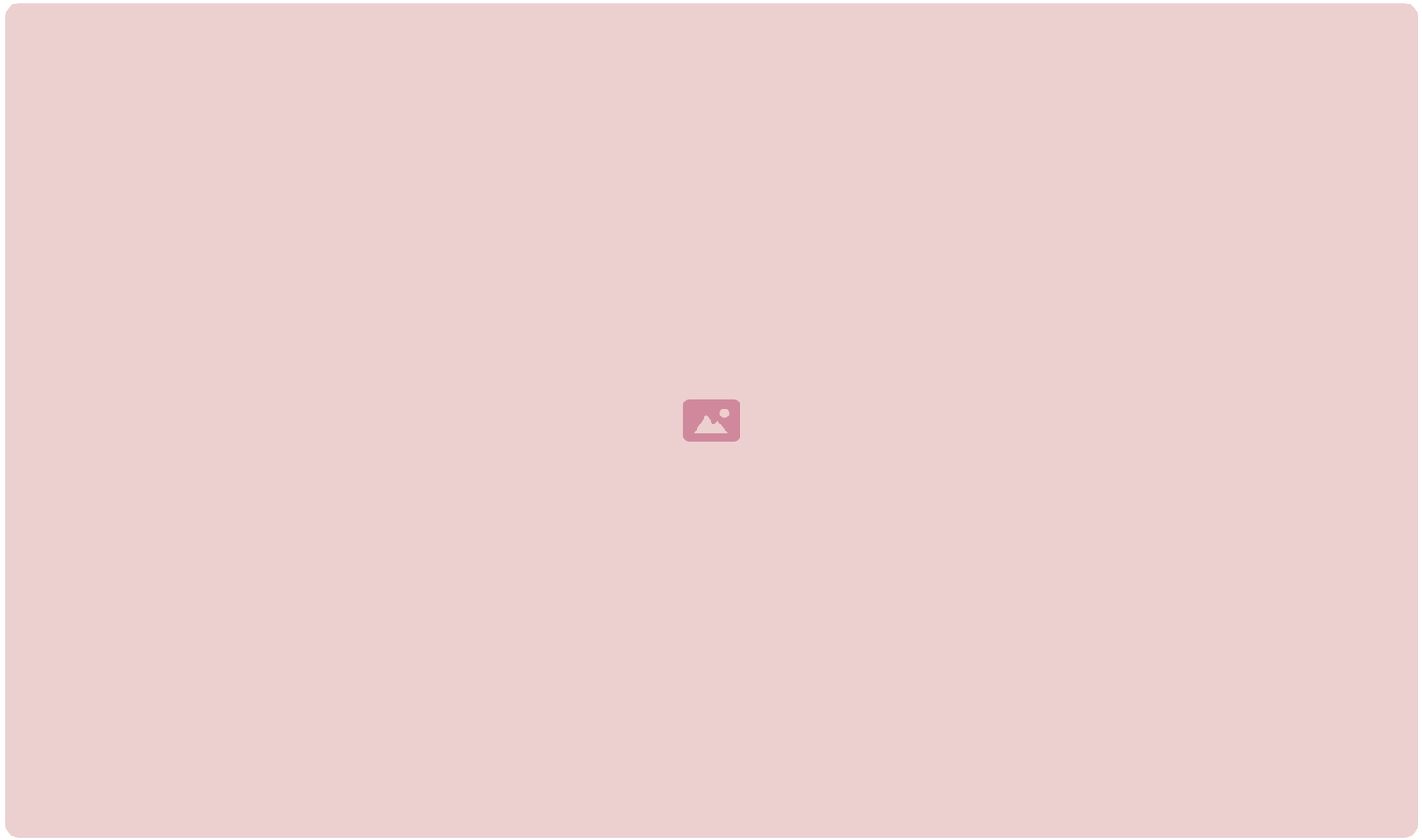
Dashboard answering both questions from gold

④ TRANSFORM + ORCHESTRATION

dbt + Prefect · extract → land → load → transform/test · every weekday 6am · retries · logged

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What the data says



STREAMLIT DASHBOARD

- Cluster buys: [top tickers + # of insiders buying]
- Insider buys vs. market: [avg. excess return]
- [Key takeaway in one line]

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Prefect run history

SCHEDULED DEPLOYMENT

- Cron: every weekday, 6am Central
- Flow: extract → land → load → dbt build
- Retries on SEC/Yahoo network calls
- [# successful scheduled runs]
- [1 retry + 1 deliberate failure test]

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06 · THE HONEST SLIDE

What broke and what I'd do next

WHAT BROKE

- [e.g. SEC requests blocked without a User-Agent]
- [e.g. ~20 GB backfill: slow, heavy first load]
- [e.g. Yahoo price gaps and rate limits]
- [Your biggest bug and how you fixed it]

WHAT I'D DO NEXT

- [e.g. Add more data tests on gold]
- [e.g. Alerting on failed runs]
- [e.g. Better benchmark for “beat the market”]
- [Your next step]

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